

HCTD: A CNN-transformer hybrid for precise object detection in UAV aerial imagery

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ABSTRACT

Object detection in UAV imagery poses substantial challenges due to severe object scale variation, dense distributions of small objects, complex backgrounds, and arbitrary orientations. These factors, compounded by high inter-class similarity and large intra-class variation caused by multi-scale targets, occlusion, and environmental interference, make aerial object detection fundamentally different from conventional scenes. Existing methods often struggle to capture global semantic information effectively and tend to overlook critical issues such as feature loss during downsampling, information redundancy, and inconsistency in cross-level feature interactions. To address these limitations, this paper proposes a hybrid CNN-Transformer-based detector, termed HCTD, specifically designed for UAV image analysis. The proposed framework integrates three novel modules: (1) a Feature Filtering Module (FFM) that enhances discriminative responses and suppresses background noise through dual global pooling (max and average) strategies; (2) a Convolutional Additive Self-attention Feature Interaction (CASFI) module that replaces dot-product attention with a lightweight additive fusion of spatial and channel interactions, enabling efficient global context modeling at reduced computational cost; and (3) a Global Context Flow Feature Pyramid Network (GC2FPN) that facilitates multi-scale semantic propagation and alignment to improve small-object detection robustness. Extensive experiments on the VisDrone2019 dataset demonstrate that HCTD-R18 and HCTD-R50 achieve 38.2%/43.7% AP₅₀, 23.1%/24.6% AP₇₅, and 13.9%/14.7% AP_S respectively. Additionally, the TIDE toolkit is employed to analyze the absolute and relative contributions of six error types, providing deeper insight into the effectiveness of each module and offering valuable guidance for future improvements. The code is available at: <https://github.com/Mundane-X/HCTD>.

1. Introduction

With the continuous reduction in UAV manufacturing costs and the rapid advancement of flight control technologies, unmanned aerial vehicles (UAVs) have been increasingly adopted in a variety of fields, such as geospatial exploration, traffic monitoring, and precision agriculture (Gao et al., 2023). These applications highlight the vast potential of UAVs in real-world scenarios. In most of these tasks, object detection serves as a fundamental and indispensable component, playing a critical role in enabling autonomous perception and decision-making. However, due to the unique flight altitudes and imaging perspectives of UAVs, aerial images often contain a large number of small objects embedded in complex backgrounds, which presents significant challenges to conventional object detection algorithms. Achieving high-precision detection under such conditions has become a core technical

bottleneck in UAV-based visual perception, and addressing this issue is essential for advancing UAV intelligence and promoting their practical deployment.

Modern object detectors have achieved remarkable success in general-purpose detection tasks; nonetheless, their performance tends to deteriorate significantly when dealing with small objects. This is mainly because small objects occupy only a few pixels in an image and are highly susceptible to feature loss during downsampling, as well as interference from background noise. As a result, detectors often fail to capture the discriminative cues needed to reliably recognize small targets in complex scenes. One of the key reasons for this performance gap is that conventional feature extraction and fusion methods are not specifically optimized for small object representation. While multi-scale architectures such as Feature Pyramid Networks (FPN) have

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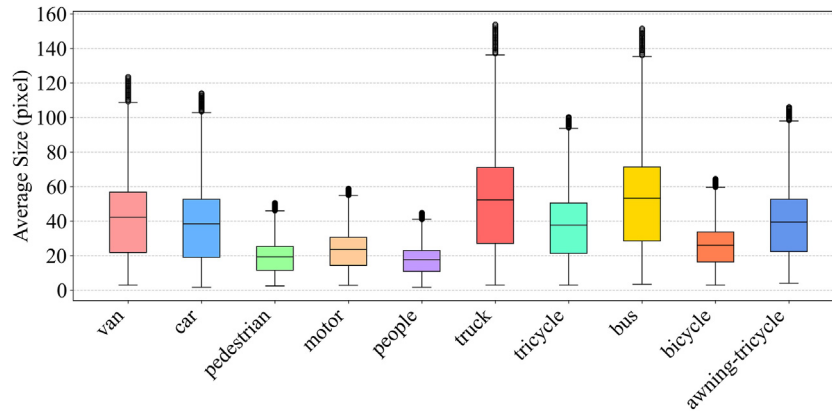


Fig. 1. Box plot of scale distribution for VisDrone2019-DET dataset.

been widely adopted to integrate low-level details with high-level semantics, naive fusion strategies often lead to feature redundancy and semantic misalignment across different scales. The former overwhelms the model with repetitive or irrelevant features, while the latter hinders the effective alignment of feature representations at various pyramid levels. In response, some recent methods attempt to improve performance by employing deeper backbones, additional network layers, or sophisticated attention mechanisms. However, these approaches generally incur substantial computational costs, which reduce inference speed and hinder real-time applications. In summary, existing methods often suffer from redundant feature information, cross-level semantic inconsistency, and high computational complexity, thus limiting their practical usability in UAV scenarios.

The challenges are further exacerbated by the unique nature of drone-captured imagery. Objects in UAV images exhibit a wide range of scales, with large structures (e.g., buildings) and tiny entities (e.g., pedestrians or vehicles) often appearing within the same frame. Additionally, UAVs capture scenes from a top-down perspective, where small objects may occupy less than 1% of the total image area and appear at arbitrary rotations, significantly altering their visual shape. As illustrated in Fig. 1, the VisDrone2019 dataset (Zhu et al., 2021) demonstrates a large volume of small objects (less than 30 pixels in size) with significant scale variations. These characteristics further confirm that UAV-specific imaging conditions adversely affect object scale and distribution, resulting in reduced detection accuracy. Moreover, as shown in Fig. 2, UAV datasets exhibit high inter-class similarity (e.g., between pedestrians and cyclists) and large intra-class variation due to factors such as flight altitude, lighting, and occlusion. These factors make object categories less distinguishable and amplify the challenge of detection. Therefore, capturing accurate semantic information becomes crucial to enhancing object discriminability and detection robustness in aerial scenarios.

To address these challenges, this paper draws inspiration from the advanced architectural principles of CSPNet (Wang et al., 2020a) and RT-DETR (Zhao et al., 2024), and leverages the superior small-object modeling capability of feature pyramid networks. We propose a novel end-to-end hybrid CNN-Transformer-based detector named HCTD. This framework is designed to enhance small object detection performance, improve real-time inference speed, and ensure deployment adaptability. The main contributions of this work are as follows:

(1) A Feature Filtering Module (FFM) is introduced to enhance salient feature responses while suppressing background noise. This module improves the perceptual saliency of target regions and aggregates global semantic information through two pooling strategies, effectively reducing feature redundancy and mitigating information loss.

(2) An efficient attention mechanism, termed Conv Additive Self-attention Feature Interaction (CASFI), is developed to replace conven-

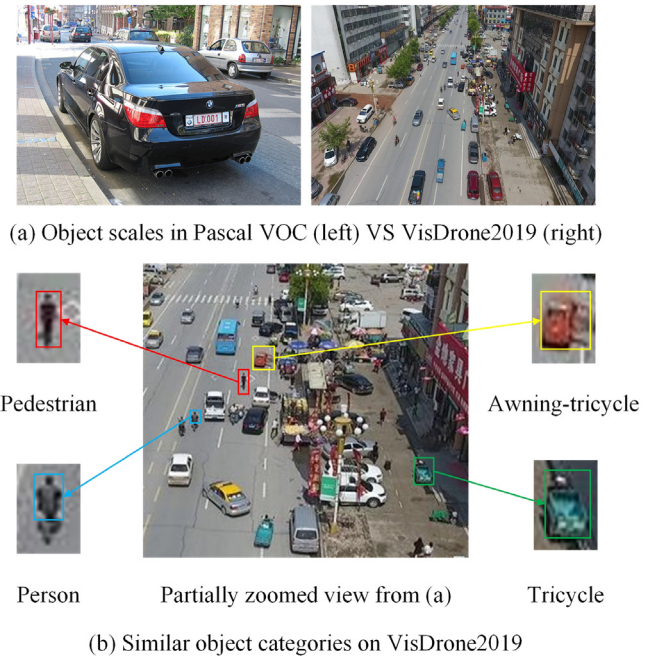


Fig. 2. The Challenge of Object Detection in VisDrone2019-DET.

tional computation-intensive attention modules. CASFI applies spatial and channel-wise attention separately to query and key features and fuses them additively. This design enables expressive token interaction with significantly lower computational overhead, enhancing the feature representation of small objects while maintaining high inference efficiency.

(3) A Global Context Flow Feature Pyramid Network (GC2FPN) is constructed to address semantic inconsistency across feature levels. This structure propagates top-level semantic information across the pyramid and facilitates horizontal context interaction within each scale. As a result, semantic consistency is improved across resolutions, leading to more robust multi-scale small object detection.

The remainder of this paper is organized as follows. Section 2 reviews related work in the field. Section 3 presents the proposed HCTD detector in detail. Section 4 reports the experimental results and comparisons. Section 5 discusses the limitations and outlines future research directions. Section 6 concludes the paper.

2. Related work

2.1. Small object detection in UAV aerial images

For small object detection in drone-captured images, Yang et al. proposed the ClusDet model (Yang et al., 2019), which adopts a global-to-local approach. First, CPNet and ScaleNet are used to identify object clusters with high aggregation in the image. Then, DetecNet performs a secondary search within these clusters. This divide-and-conquer approach establishes a general research paradigm. Li et al. introduced DMNet (Li et al., 2020), which reduces the model complexity of ClusDet by using pixel intensity-guided learning of class ratios. DMNet emphasizes the spatial relationships between objects through its composite architecture, which includes a density map generation module, image cropping module, and object detector. It achieved strong performance on aerial image datasets. The UFPMP-Det model proposed by Huang et al. (2022) involves two key processing steps. First, the Unified Foreground Packing module merges the sub-regions provided by the coarse detector to reduce background interference. Then, the Multi-Proxy Detection Network models the objects in a fine-grained manner. The combination of these two methods strikes a balance between speed and accuracy. GLSAN (Deng et al., 2020) and CRENet (Wang et al., 2020b) improve the clustering prediction algorithm by sending more accurate object aggregation regions into the fine detector, thereby enhancing detection performance. Although the aforementioned methods can detect small-sized objects, their performance deteriorates when dealing with images containing dense small objects. To address this, Zhang et al. proposed SGMFNet (Zhang et al., 2023), which includes a global-local feature guidance module that effectively integrates global and local information, reducing interference from complex backgrounds. Additionally, an inverse-residual feature enhancement module is added before the detection head to improve the extraction of small object features. The model demonstrates significant improvements on the VisDrone2019 dataset. The CEASC model proposed by Du et al. (2023) is specifically optimized for resource-constrained drone platforms. It improves the detection head using sparse convolutions and employs an adaptive multi-layer masking strategy to generate optimal mask ratios on feature maps at different scales, achieving compact foreground coverage to improve accuracy and efficiency. Xi et al. introduced FiFoNet (Xi et al., 2022), which effectively selects multi-scale features and background interference combinations for the object, further restores the differentiability of multi-scale features, and enhances the representation of small objects with low-quality features, resulting in performance improvements across multiple drone image datasets. Zhu et al. (2024) presented a method that integrates global multi-level perception with dynamic region aggregation, leveraging self-attention to capture global context and a sparse selection strategy to aggregate key local features. This approach effectively reduces background noise and enhances fine-grained details, achieving strong performance on densely distributed small targets across several benchmark datasets. Li et al. (2024b) introduced a detection framework based on a global and local attention mechanism (GAL) for remote sensing imagery, combining fine-grained local feature modeling with global semantic information. Their model further incorporates multi-scale decoupled prediction heads and a dedicated small-object localization loss, resulting in notable improvements on challenging datasets such as VisDrone2019 and DOTA. Qi et al. (2022) proposed a single-stage detector, SODNet, which employs an Adaptive Spatial Parallel Convolution (ASPCov) module to extract multi-scale spatial features, and a Fast Multi-Scale Fusion (FMF) module to efficiently aggregate semantic and spatial information. Their design enhances small-object representation while maintaining real-time inference speed.

Inspired by the aforementioned methods, this work also integrates multi-level feature fusion and attention mechanisms, but places greater emphasis on computational efficiency and deployment flexibility. A

conv additive self-attention feature interaction module (CASFI) is proposed to replace the computationally intensive self-attention mechanism with an efficient convolutional additive token interaction design. In addition, a Feature Filtering Module (FFM) is introduced to suppress redundant noise, and a Global Context Flow Feature Pyramid Network (GC2FPN) is designed to enable cross-layer semantic fusion, thereby enhancing semantic consistency for small object representations. The overall architecture is optimized for resource efficiency, maintaining competitive accuracy while achieving strong real-time performance and deployment adaptability.

2.2. Feature pyramid network

Feature pyramids are a technique used to handle multi-scale features and improve object detection accuracy. The core goal is to process feature maps at different levels, enabling the model to perform well on objects of various sizes. Based on traditional image pyramid methods, FPN (Lin et al., 2017) optimizes and innovates by introducing a top-down feature propagation mechanism and lateral connections. This combines high-level semantic features with low-level spatial features, avoiding information loss inherent in traditional methods. FPN improves detection of small objects while retaining the accuracy of large object detection. PANnet (Liu et al., 2018) builds upon FPN by introducing a path aggregation mechanism, which enhances the flow of context information from low-level to high-level features and ensures interaction between low-level and high-level feature maps. However, while PANnet improves performance, it also increases computational overhead, leading to longer training and inference times, making real-time inference challenging on hardware with limited computational resources. Google proposed an enhanced Feature Pyramid Network (BiFPN) (Tan et al., 2020) to further improve the efficiency and effectiveness of multi-scale feature fusion in object detection models. BiFPN introduces a bidirectional feature fusion mechanism that effectively integrates features from both low and high layers. Additionally, it employs weighted feature fusion, assigning different weights to each layer, which enhances the effectiveness of information transmission. To improve computational efficiency, EfficientRepGFPN (Xu et al., 2022) incorporates a repeated group feature aggregation mechanism, further strengthening feature fusion through multiple aggregation operations. This method enables more effective combination of features from different scales and optimizes computation through grouping, reducing redundant calculations. RepPAN (Li et al., 2022) is an improved feature pyramid network that introduces repeated path aggregation, allowing each feature map to pass information through different paths, ensuring the full utilization of feature complementarity between high and low layers. Its adaptive feature fusion mechanism automatically assigns weights based on the importance of features from different scales, achieving more precise feature fusion. AFPN (Yang et al., 2023) overcomes the limitations of traditional FPN in handling multi-scale features by adaptively adjusting and optimizing the feature fusion process. It dynamically selects the most suitable feature fusion method based on the different scales of features and objects in the image. For small objects, it relies more on low-level features, while for large objects, it utilizes high-level features more effectively.

Although the aforementioned feature pyramid networks hold significant research value, most of them are designed based on conventional scene images. While some methods are effective in aerial image object detection, they do not specifically account for the positional information of objects and global context in aerial images, thereby limiting the detection performance.

2.3. Vision transformer & DETRs

ViT (Dosovitskiy, 2020) is a revolutionary vision model based on Transformer (Vaswani, 2017), introducing a new paradigm to the field of computer vision and altering traditional approaches to image

processing and understanding. Due to the high computational complexity of ViT, many subsequent studies have focused on lightweight solutions. Swin Transformer (Liu et al., 2021) introduces a sliding window technique and a local window attention mechanism, which not only improves computational efficiency but also retains the global modeling advantages of Transformer. Its hierarchical structure enables the network to capture information at different scales, enhancing efficiency and enabling the processing of larger images. To further optimize self-attention techniques, EfficientViT (Liu et al., 2023) employs local window self-attention, where self-attention is computed within each window, reducing computational complexity and improving efficiency. The outputs from multiple windows can capture global information through sliding or cross-window connections, thus enhancing the model's performance.

The Vision Transformer (ViT) was originally designed for image classification tasks, and Facebook built DETR (Carion et al., 2020) on top of it, specifically optimized for object detection tasks. DETR simplifies the traditional object detection pipeline into an end-to-end process, eliminating the need for steps such as proposal generation, classification regression, and non-maximum suppression, offering a novel approach to object detection. DETR demonstrates significant advantages in capturing global context information in images and handling multi-object scenarios with complex backgrounds. However, it suffers from slow training and inference speeds, prompting subsequent research to focus on improving its efficiency. Deformable-DETR (Zhu et al., 2020) introduces a deformable attention mechanism, which selectively attends to more important regions in the image while ignoring irrelevant background areas. Additionally, the dynamic query mechanism and improved encoder-decoder architecture facilitate more efficient interaction of image features, resulting in strong performance across various scenarios. Efficient-DETR (Yao et al., 2021) is an efficient improvement of DETR, designed to enhance training speed, inference speed, and computational efficiency for object detection tasks. By employing efficient multi-scale feature extraction, an improved self-attention mechanism, and enhanced decoder structures, Efficient-DETR achieves comparable detection accuracy to DETR but with higher efficiency.

ViTs and DETRs have introduced remarkable global semantic information synergy in image feature representation, but this comes at a high computational cost. At the same time, they overly focus on global features while neglecting local features, which results in poor detection performance for small objects.

3. Methodology

The overall structure of HCTD is shown in Fig. 3, consisting of the backbone, hybrid encoder, and hybrid decoder with a prediction head. Specifically, the feature selection module filters the features from the last four stages of the backbone (S_2, S_3, S_4, S_5) to obtain refined features (P_2, P_3, P_4, P_5). The hybrid encoder converts the multi-scale features into a series of image features through Convolutional additive self-attention and the global context flow feature pyramid network. Then, query selection is applied to choose a fixed number of encoded features as the initial object queries for the decoder. Finally, the decoder, equipped with a prediction head, iteratively refines the object queries to generate class labels and bounding boxes.

3.1. Feature filtering module

The feature filtering module uses Context Select Attention (CSA) as the core structure. The CSA module processes the input features using global average pooling and global max pooling, respectively. The combination of these two operations allows the model to retain the strongest response values in the feature map, emphasize the salient regions in the image, reduce the impact of noise on the model, and simultaneously aggregate the average values to capture global semantic information, minimizing information loss. The pooled information is

then fed into horizontal depth-wise strip convolution blocks, vertical depth-wise strip convolution blocks, and depth-wise convolution blocks to capture long-range directional features and integrate inter-channel dependencies, enhancing the feature representation capability. The depth-wise strip convolutions operate in horizontal and vertical directions, utilizing larger kernels to capture global contextual information in both directions, with half the computational complexity of conventional convolution kernels. However, this method can only capture contextual information in a single direction, which is insufficient for complex or multi-directional object shapes. To address this limitation, we introduce depth-wise convolution, which independently extracts spatial features via depth-wise convolution, followed by point-wise convolution to integrate channel information. The all-direction receptive field of this structure removes directional constraints, enabling the capture of global context and the generation of a mixed attention map, further improving the feature representation capability. Finally, the feature information generated by the two convolution blocks is concatenated along the channel dimension and passed through a 1×1 convolution followed by a Sigmoid activation function to generate the attention weights. The structure of the feature filtering module is shown in Fig. 4.

3.2. Conv additive self-attention feature interaction

The introduction of the Vision Transformer (ViT) marks a revolutionary shift in the architecture of computer vision models. ViT employs a novel approach called the Token Mixer (Huang et al., 2023), which captures long-range dependencies between features and exhibits strong global modeling and representation capabilities. The basic Token Mixer primarily consists of a multi-head self-attention mechanism (Yu et al., 2022), which compensates for the limitations of CNNs but comes with higher computational complexity, making it unsuitable for real-time applications. Therefore, we use Convolutional Additive Self-attention (Zhang et al., 2024) to improve the multi-head self-attention mechanism and construct a Convolutional Additive Token Mixer, as shown in Fig. 5. The similarity function between $Q \in R^{N \times D}$ and $K \in R^{N \times D}$ is defined as:

$$\text{Sim}(Q, K) = \Phi(Q) + \Phi(K) \quad (1)$$

$$\Phi(Q) = C(S(Q)) \quad (2)$$

$$\Phi(K) = C(S(K)) \quad (3)$$

In the equation, Q and K are obtained through independent linear transformations. The function Sim denotes a similarity modeling operation that replaces the dot-product attention mechanism in traditional Transformers. It extracts meaningful information from the query and key vectors using spatial and channel attention mechanisms, respectively, and then fuses them additively. This design enables efficient and expressive token interaction without incurring the high computational cost of matrix multiplication. The function Φ is defined as a composite attention operator that performs two consecutive operations, encapsulating the essential information interaction. The spatial attention operator $S(\cdot)$ integrates depthwise separable convolution, batch normalization, and Sigmoid activation to effectively capture spatially salient regions. The channel attention operator $C(\cdot)$ employs global average pooling followed by a 1×1 convolution and Sigmoid activation to model inter-channel dependencies. Unlike conventional Transformer architectures that incorporate explicit positional encoding — such as sinusoidal functions or learnable embeddings — to provide spatial information, our proposed design omits external position encoding. Instead, we rely on the inherent inductive biases of convolutional operations to implicitly model spatial relationships. Specifically, the spatial attention operator $S(\cdot)$ employs depthwise separable convolutions with a 3×3 kernel, enabling the network to capture local spatial context through the convolutional receptive field. This mechanism effectively

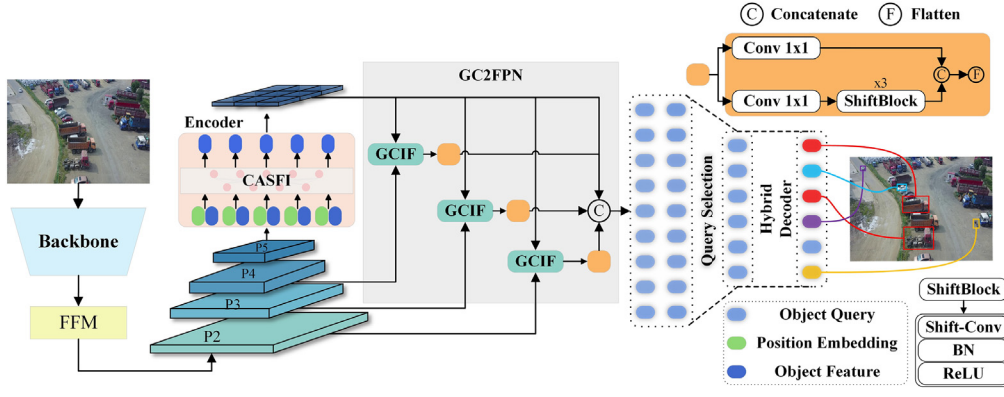
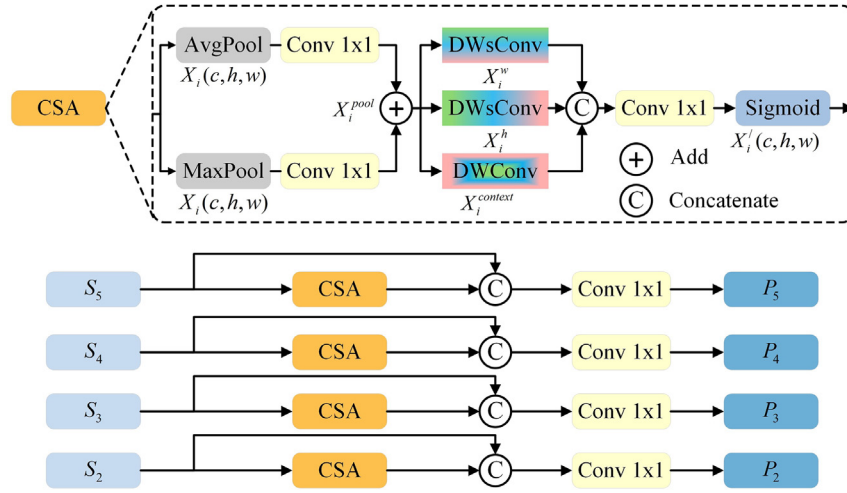


Fig. 3. The overall architecture of HCTD.

Fig. 4. Feature Filtering Module (FFM). DWsConv represents Depth-wise strip Convolution, DWConv represents Depth-wise Convolution, S_i represents the input features from the backbone, and P_i represents the output features from the feature filtering module.

embeds relative positional information within the feature maps, enhancing the model's ability to distinguish spatially variant patterns without introducing additional positional embeddings. As a result, the model benefits from efficient spatial encoding while maintaining low computational overhead.

3.3. Global context flow feature pyramid network

In the VisDrone2019 dataset, the object detection task faces the challenge of multi-scale issues, which makes it difficult for the model to accurately recognize objects in drone-captured images. This problem arises due to the significant size differences between different types of objects, and even within the same category, objects may exhibit large size variations at different aerial viewpoints. To address the multi-scale challenge in aerial datasets, we propose a feature pyramid network with global context information flow (GC2FPN) for multi-scale feature interaction. This enables the model to capture more comprehensive features of aerial objects. GC2FPN consists of two key feature processing structures: (1) the global context semantic feature fusion module. (2) the feature enhancement module. First, the features filtered by the FFM are individually fed into hierarchical cascade channels. Then, through the Global Context Information Selection (GCIS) module, high, middle, and low-level information in the feature vectors is thoroughly mixed. The top-down information flow allows mid-to-low-level features to contain rich contextual semantic information, which helps improve the recognition of fine-grained features in aerial images, thereby enhancing the model's detection capability. Each fusion structure is followed by

a feature enhancement module, which primarily consists of Conv 1x1 and Shift-Conv (Wu et al., 2018) to adjust the feature channel number and improve the model's non-linear representation ability.

Global Context Information Fusion Module: The high-level feature maps generated by the backbone network contain rich semantic information, but the object location and detail information are relatively coarse. In contrast, low-level features provide precise object locations and texture details, but lack rich contextual semantic information. The conventional solution to this problem is to directly sum the upsampled high-level feature maps with low-level feature maps at corresponding pixels, unselectively passing high-level semantic information to each layer of the low-level feature maps. However, this approach does not filter the high-level feature information; instead, it simply adds pixel values across multiple layers, leading to the inclusion of redundant information for subsequent operations. To address this issue, we developed the Global Context Information Fusion Module (GCIF), which filters the global contextual information from the high-level features and embeds the filtered features into the low-level features, achieving selective fusion of important semantic information. As shown in Fig. 6, given a high-level input feature $I_{high} \in R^{C \times H \times W}$ and a low-level input feature $I_{low} \in R^{C \times H_1 \times W_1}$, a transposed convolution with a stride of 2 and a kernel size of 3×3 is first applied to initialize the high-level input features, yielding $I'_{high} \in R^{C \times 2H \times 2W}$. To align the dimensions of the high-level and low-level inputs, bilinear interpolation is then used to scale I'_{high} , resulting in $I^{tb}_{high} \in R^{C \times H_1 \times W_1}$. The CSA module is subsequently employed to transform the high-level features into a filtering weight matrix, which is used to filter the low-level

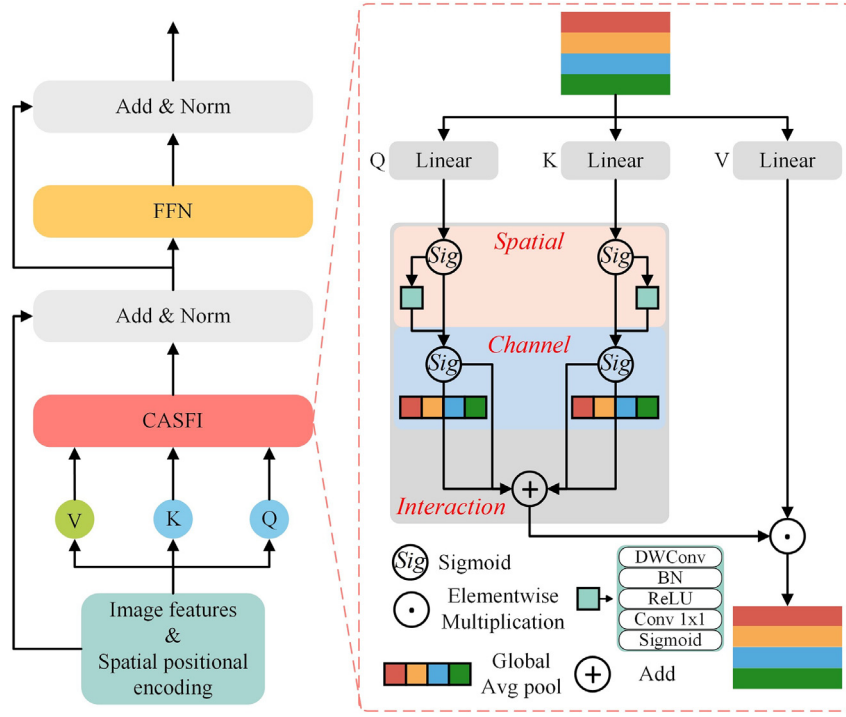


Fig. 5. Encoder and CASFI.

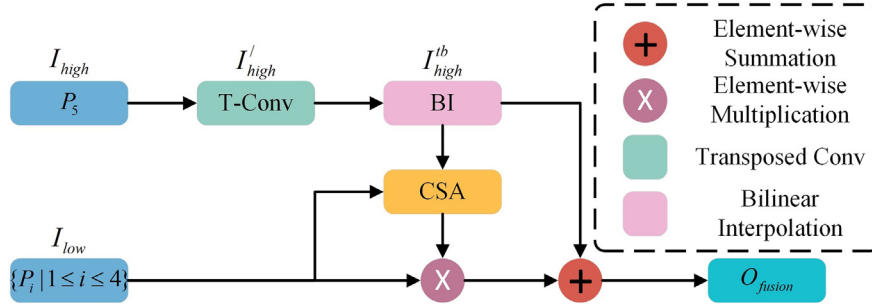


Fig. 6. The illustration of Global Context Information Fusion Module.

features. Finally, the filtered low-level features are fused with the high-level features to obtain the final output feature O_{fusion} .

During the upsampling process of high-level features, both transposed convolution and bilinear interpolation are employed to enlarge the scale of the high-level features. Bilinear interpolation is a pixel-weighted average interpolation method that does not require training or parameter learning, making it easy to implement and computationally efficient. However, it only performs a simple weighted average, which may not accurately recover complex structures or high-frequency details. Therefore, we first apply transposed convolution to process the high-level features before using bilinear interpolation. Transposed convolution learns features through convolutional kernels and generates more details during upsampling, retaining more image information while optimizing the output based on contextual information. However, transposed convolution may introduce a "checkerboard effect," where periodic stripe-like noise appears on the image, as the features learned by the convolution kernel may overemphasize details in certain regions. To mitigate this negative effect, bilinear interpolation is applied after transposed convolution.

Shift Feature Enhancement Module: Previous studies often enhanced intermediate features by combining linear layers with activation functions or by cascading several 1x1 convolutions with activation functions, while introducing non-linearity. Although this approach has

shown some effectiveness in various tasks, it still suffers from insufficient spatial information modeling and high computational memory overhead. To achieve more effective feature enhancement, we chose to use a combination of Shift-conv and ReLU activation functions. Shift-conv consists of four shift convolutions and a 1×1 convolution. Let the input feature map be $X \in R^{C \times H \times W}$, where C denotes the number of input channels, and H and W represent the width and height of the feature map. Let the convolution kernel be $K \in R^{C_{out} \times C \times H \times W}$, where C_{out} denotes the number of output channels and K is the kernel size ($K = 3$). The height and width of the output feature map can be computed using the following formula:

$$H' = \frac{H + 2p - K}{s} + 1 \quad (4)$$

$$W' = \frac{W + 2p - K}{s} + 1 \quad (5)$$

where p is the padding size and s is the step size.

In Shift-Conv, an offset mask $\Delta = \delta_{i,j}$ is introduced, which defines the offset for each channel. For the input feature map $X(h, w)$, the offset X' is given by:

$$X'(h', w') = X(h + \delta_{i,j}^h, w + \delta_{i,j}^w) \quad (6)$$

where $\delta_{i,j}^h$ is the vertical offset and $\delta_{i,j}^w$ is the horizontal offset, each channel needs to be processed. During the implementation, the offset

may cause the sampling point to be outside the feature map boundary, which is supplemented by padding with 0. The convolution operation of Shift-Conv is completed based on the offset sampling points. For each position (i, j) and channel C of the convolution kernel, the calculation formula of the output feature $O(h, w)$ is as follows:

$$O(h, w) = \sum_{c=1}^C \sum_{i=1}^k \sum_{j=1}^k K(c, i, j) \cdot X'(h', w') \quad (7)$$

The key to Shift-Conv lies in altering the sampling locations of the convolution kernel through the offset mask Δ , enabling more flexible spatial information modeling. Its formulation can be seen as an extension of traditional convolution, retaining the efficiency of convolution while significantly enhancing the ability to perceive both local and global spatial features.

After performing the offset convolution, the output is normalized to ensure that each channel has zero mean and unit variance. Standard normalization removes the mean and variance of the data, stabilizing the output of each layer and accelerating the convergence process. However, normalizing the output may limit the model's representational capacity, as directly passing the normalized values to the next layer might cause the loss of critical information. To address this issue, we introduce two learnable parameters in batch normalization: the scaling parameter γ_c and the shifting parameter β_c . The γ_c parameter acts as a "scaler," adjusting the importance of the normalized features by amplifying or shrinking the feature values, while β_c functions as a "shifter," moving the feature values to a more suitable range for improved learning. These parameters are trained via backpropagation, allowing the network to automatically optimize them for different tasks. The subsequent batch normalization and ReLU activation operations are given by the following formula:

$$\hat{O}_{b,c,h,w} = \frac{O_{b,c,h,w} - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}} \quad (8)$$

$$O'_{b,c,h,w} = \gamma_c \hat{O}_{b,c,h,w} + \beta_c \quad (9)$$

$$O_{b,c,h,w}^{active} = \text{ReLU}(O'_{b,c,h,w}) \quad (10)$$

where μ_c denotes the mean of the activations in channel c , computed across the entire mini-batch and spatial dimensions. σ_c^2 denotes the variance of channel c , reflecting the fluctuation of feature values across spatial and batch dimensions within that channel. ϵ is a small constant (e.g., 10^{-5}) added for numerical stability to prevent division by zero during standardization.

4. Experiments

4.1. Dataset

The VisDrone2019 dataset is one of the leading drone-based aerial imagery datasets, collected and developed by Tianjin University in collaboration with the data mining team AISKYEYE. The dataset consists of images recorded by various cameras mounted on drones, covering different locations (14 cities across China), environments (urban and rural areas), objects (pedestrians, vehicles, bicycles), and scene densities (sparse and crowded scenarios). It includes 10 object categories frequently encountered in drone applications. The dataset contains a total of 7019 images, with 6471 images used for training and 548 images for validation. Object instance bounding boxes for these categories are annotated within the dataset. Additionally, important attributes such as scene visibility, object categories, and occlusion are provided to better utilize the data for research purposes.

4.2. Implementation details

We constructed HCTD using the PyTorch deep learning framework, the MMDetection Toolbox, and the Ultralytics Toolbox, and performed training and inference on two NVIDIA Tesla V100 GPUs. HCTD uses the same input size $640 \times 640 \times 3$ as YOLO, RCNN, and RT-DETR, while other DETR detectors use an input size of 800×1333 . The batch size is set to 8. To ensure fairness and effectiveness in the experiments, no manual parameter tuning was performed for any of the models. Except for the YOLO series, all models use ResNet50 (R-50) and ResNet18 (R-18) as baseline feature extraction networks, and all models were implemented on the same hardware. All metrics in this paper are computed based on the VisDrone2019 test set, with 10 categories. The experiments use the conventional mAP metric and the COCO AP metric as key performance indicators for detection, where mAP is calculated based on the average over 10 IoU thresholds and all categories. COCO also provides evaluation metrics at different IoU thresholds and for objects of various sizes. Specifically, AP_S is used to evaluate small objects with pixel areas smaller than 32×32 , AP_M is used for medium objects with pixel areas between 32×32 and 96×96 , and AP_L is used for large objects with pixel areas greater than 96×96 . AP_{50} and AP_{75} are evaluation metrics for models at IoU thresholds of 50% and 75%, respectively. Additionally, we use the Tide (Bolya et al., 2020) tool for a more comprehensive analysis of common errors in object detection. The model scale for all experiments is primarily determined by the number of parameters.

4.3. Comparison with advanced detectors

In this experiment, we performed a detailed comparison of the current state-of-the-art object detection models, with a particular focus on evaluating the performance of our proposed HCTD model in terms of accuracy, speed, and multi-scale object detection. The experimental results, as shown in the table, demonstrate that HCTD excels in both detection performance and inference speed, particularly showing significant advantages in efficient real-time detection scenarios. Table 1 presents the results of the comparison experiments.

In terms of accuracy metrics, HCTD performs exceptionally well across different evaluation standards. With the ResNet-50 backbone network, the HCTD model achieves an AP_{50} of 43.7, surpassing other state-of-the-art models such as RT-DETR-R50 (38.5) and DINO-DETR-R50 (43.4). In the more challenging AP_{75} metric, HCTD-R50 reaches 24.6, which is close to DINO-DETR-R50 (24.7), but with a significant advantage in inference speed. Furthermore, HCTD-R50 achieves 14.7 and 37.8 for small and medium-scale object detection (AP_S and AP_M), respectively, matching the performance of the best DINO-DETR-R50. For large-scale object detection (AP_L), HCTD-R50 is nearly on par with DINO-DETR-R50 (50.8), while significantly outperforming other models such as YOLOv5-L (40.7), ATSS-R50 (40.9), RetinaNet-R50 (42.3), TOOD-R50 (43.0), Faster-RCNN-R50 (43.0), RT-DETR-R50 (43.4), YOLOv8-L (45.6), and YOLOv10-X (46.6), demonstrating excellent adaptability to objects of various scales. With the ResNet-18 backbone, the HCTD model outperforms YOLOv5-M, YOLOv8-M, YOLOv10-L, ATSS-R18, RetinaNet-R18, TOOD-R18, Faster-RCNN-R18, DINO-DETR-R18, and RT-DETR-R18, achieving improvements in AP_{50} of 26.9%, 10.1%, 2.4%, 7.6%, 11.4%, 29.5%, 3.5%, 0.3%, and 4.7%, respectively. For more challenging metrics such as AP_{75} and AP_S , HCTD-R18 continues to lead and achieves the best results. This demonstrates that our proposed model maintains strong performance even when using different backbone networks. Fig. 7 presents a comparison of detection accuracy across 10 categories between HCTD with two different backbone sizes and other models.

In terms of inference speed, HCTD stands out significantly. With the ResNet-50 backbone, the HCTD model achieves an inference speed of 94 FPS, significantly outperforming high-precision models such as

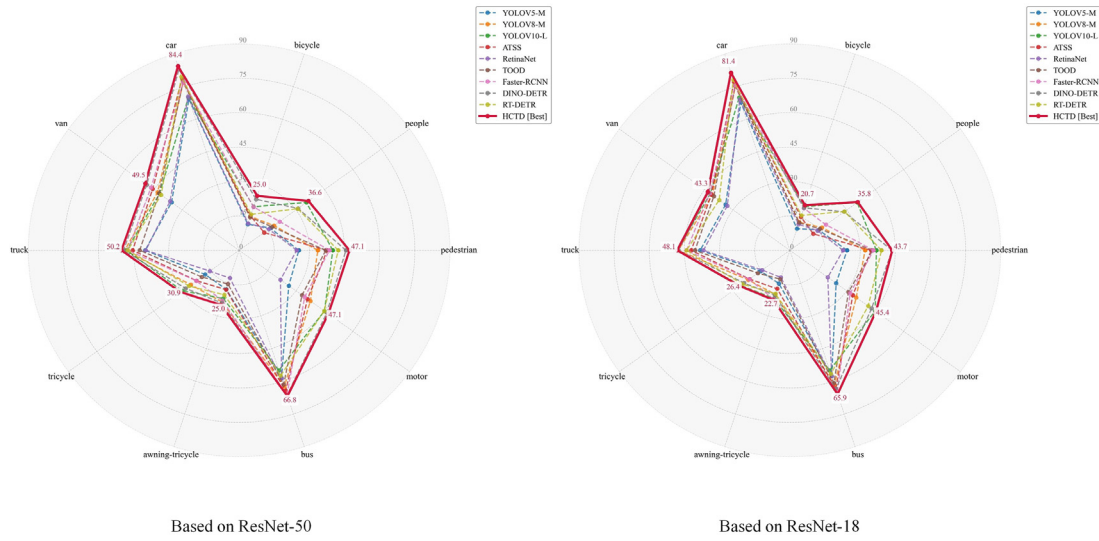


Fig. 7. Radar chart of detection accuracy for each object class across different models.

Table 1
Comparison with SOTA detectors.

Model	Backbone	Params(M)	FPS _{bs=1}	AP ₅₀ (%)	AP ₇₅ (%)	AP _S (%)	AP _M (%)	AP _L (%)
YOLOv5-M (Jocher et al., 2022)	–	21.2	63	30.1	18.3	7.7	27.6	38.3
YOLOv5-L (Jocher et al., 2022)	–	46.5	55	31.9	19.6	8.3	29.7	40.7
YOLOv8-M (Xue et al., 2024)	–	25.8	85	34.7	20.4	9.8	30.7	42.7
YOLOv8-L (Xue et al., 2024)	–	43.2	72	36.9	22.0	10.6	33.0	45.6
YOLOv10-L (Wang et al., 2024)	–	24.4	126	35.8	20.7	10.1	30.8	43.4
YOLOv10-X (Wang et al., 2024)	–	29.5	95	38.0	22.4	11.5	33.3	46.6
ATSS (Zhang et al., 2020)	R-18	19.0	–	35.5	20.5	12.0	30.7	35.3
ATSS (Zhang et al., 2020)	R-50	31.9	–	40.2	24.6	14.3	35.7	40.9
RetinaNet (Ross and Dollár, 2017)	R-18	19.8	–	34.3	19.2	9.7	30.9	37.9
RetinaNet (Ross and Dollár, 2017)	R-50	39.3	–	38.7	23.0	11.5	36.2	42.3
TOOD (Feng et al., 2021)	R-18	18.9	49	29.5	18.3	8.1	26.1	37.2
TOOD (Feng et al., 2021)	R-50	31.8	30	32.2	20.5	9.5	30.3	43.0
Faster-RCNN (Ren et al., 2016)	R-18	28.2	47	36.7	22.5	13.7	32.0	37.3
Faster-RCNN (Ren et al., 2016)	R-50	41.2	28	42.4	24.4	15.6	36.5	43.0
DINO-DETR (Zhang et al., 2022)	R-18	25.3	–	38.1	22.9	13.7	36.0	38.1
DINO-DETR (Zhang et al., 2022)	R-50	47.5	–	43.4	24.7	14.6	37.5	50.8
RT-DETR (Zhao et al., 2024)	R-18	20.1	136	36.5	20.9	11.6	31.0	31.3
RT-DETR (Zhao et al., 2024)	R-50	42.3	98	38.5	22.4	12.7	32.5	43.4
HCTD	R-18	23.6	120	38.2	23.1	13.9	35.9	38.0
HCTD	R-50	45.4	94	43.7	24.6	14.7	37.8	50.6

Faster-RCNN-R50 (28 FPS) and DINO-DETR-R50. Compared to real-time detectors such as the YOLO series, HCTD-R50 also performs exceptionally well, achieving FPS improvements of 70.9% and 30.6% over YOLOv5 and YOLOv8, respectively. When compared to the fastest real-time models, YOLOv10-X and RT-DETR-R50, HCTD-R50 has a minimal average gap of only 2.65%, while surpassing them in accuracy by 14.25%, which is an acceptable trade-off. The lightweight HCTD model with the ResNet-18 backbone achieves an outstanding 120 FPS while maintaining excellent accuracy. Compared to YOLOv5, YOLOv8, TOOD-R18, and Faster-RCNN-R18, the FPS improves by 90.5%, 41.2%, 144.9%, and 155.3%, respectively.

In terms of parameter scale, the HCTD-R50 model has 45.4M parameters, which is close to YOLOv5-L (46.5M) and DINO-DETR-R50 (47.2M), but HCTD exhibits clear advantages in both detection speed and accuracy. The HCTD-R18 (23.6M) model, compared to larger models such as YOLOv8-M (25.8M), YOLOv10-L (24.4M), and Faster-RCNN-R18 (28.2M), demonstrates a comprehensive improvement in both detection accuracy and speed. Fig. 8 presents the visualized detection results of HCTD under various crowding levels and shooting angles.

4.4. Detection performance comparison based on feature pyramids

To thoroughly evaluate the performance of GC2FPN in object detection tasks, we conducted a comparative experiment on the VisDrone2019 test set, comparing GC2FPN with various advanced feature pyramid networks based on RT-DETR. The focus was on analyzing the performance of the proposed GC2FPN feature pyramid structure. The results are shown in Table 2. Our GC2FPN achieved excellent results across RT-DETR models with different backbone feature extraction networks.

When using ResNet-18 as the backbone, GC2FPN achieves an AP₅₀ of 37.4%, significantly outperforming standard FPN (35.4%), SlimNeck (36.4%), AugFPN (36.5%), and BiFPN (36.9%). For the higher precision metric, AP₇₅, GC2FPN reaches 22.2%, comparable to CFPT and outperforming other models. With ResNet-50 as the backbone, GC2FPN achieves an AP₅₀ of 40.9%, surpassing SlimNeck (38.6%), BiFPN (39.0%), and SSFPN (39.9%), achieving the highest performance among all compared methods. The AP₇₅ for GC2FPN is 23.2%, again surpassing other models such as CFPT (23.0%) and AugFPN (22.5%), demonstrating its excellent high-precision detection capability. Small object detection in aerial images is a challenging task in

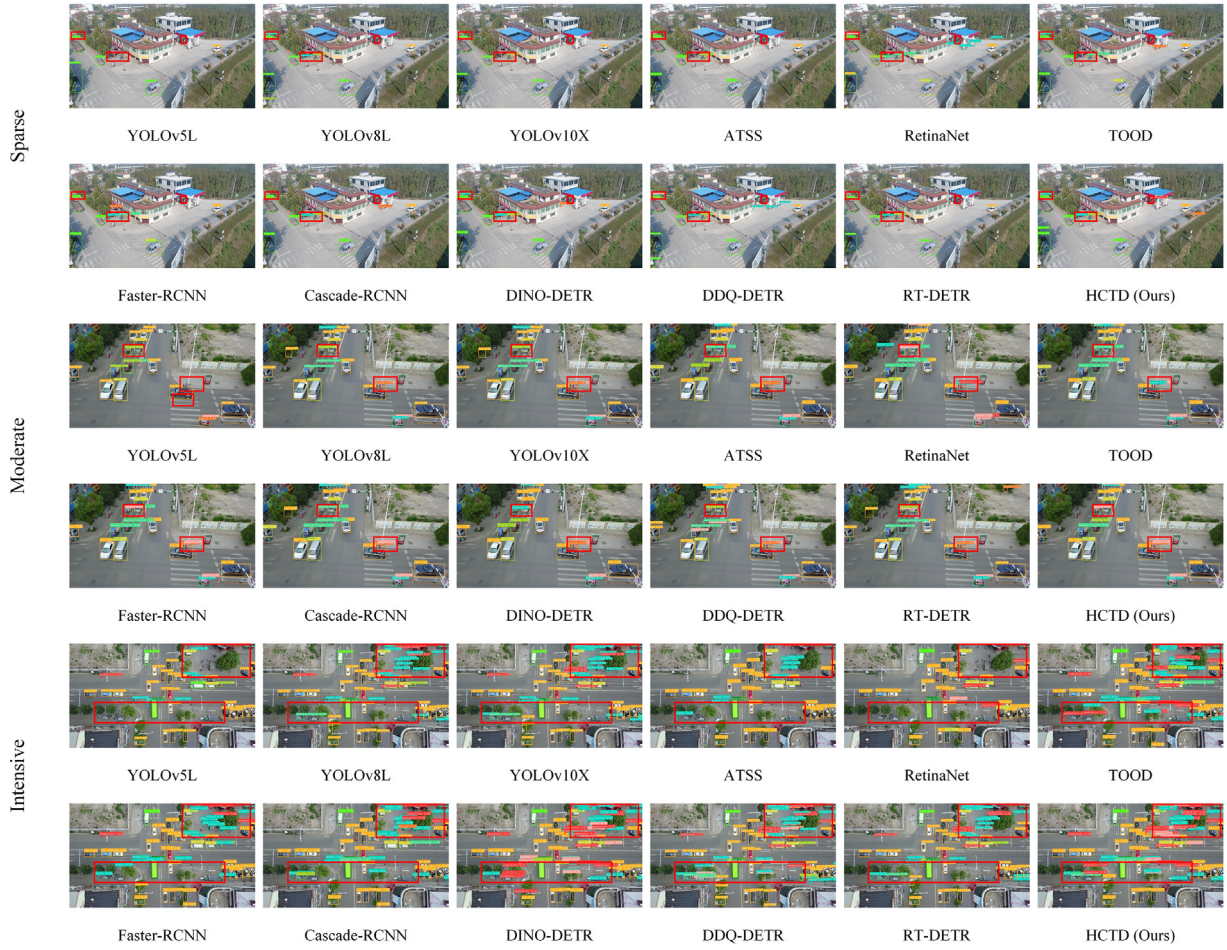


Fig. 8. The visualization of detection results for different models, with regions showing significant differences highlighted using red bounding boxes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

object detection. When using ResNet-18 and ResNet-50 as backbones, GC2FPN achieves AP_S of 12.3% and 13.8%, respectively, outperforming all other models, demonstrating higher sensitivity to small-sized objects in aerial images. In the medium-sized object detection task, with ResNet-18 as the backbone, GC2FPN (32.9%) outperforms most models, including FPN (30.4%), BiFPN (31.7%), and SlimNeck (32.0%). With ResNet-50 as the backbone, GC2FPN achieves an AP_M of 35.6%, significantly outperforming the base FPN (30.4%) and surpassing other advanced feature pyramid structures. Although GC2FPN's AP_M with ResNet-18 and ResNet-50 is slightly lower than CFPT, the gap is within 1%, indicating that GC2FPN is competitive with CFPT. In the large object detection task, with ResNet-18 as the backbone, GC2FPN achieves an AP_L of 35.7%, clearly surpassing SlimNeck (31.4%), AFPN (32.8%), and RCFPN (33.6%). With ResNet-50 as the backbone, GC2FPN reaches an AP_L of 45.8%, well ahead of SlimNeck (43.9%), AFPN (44.4%), AugFPN (44.7%), and RCFPN (44.9%), demonstrating its strong capability in large object detection. Through the comparative analysis above, GC2FPN demonstrates superior performance across key metrics (AP_{50} , AP_{75} , AP_S , AP_L), with particularly significant advantages in small object detection and high-precision detection, showcasing excellent multi-scale adaptability. Although its parameter count is slightly higher than that of the standard FPN, the performance improvement is notable, indicating better parameter utilization efficiency. Overall, GC2FPN achieves a good balance between detection performance and computational complexity through efficient feature fusion and optimized design, providing strong support for high-precision object detection tasks in aerial images.

4.5. Fine-grained error analysis and comparative evaluation

In the previous section, we used COCO metrics as the evaluation standard and conducted comparative experiments with various advanced detectors, demonstrating the superiority of HCTD from multiple perspectives. However, COCO's AP (Average Precision) and AR (Average Recall) only provide an overall performance evaluation and do not reveal the different types of detection errors, which are often the primary performance bottlenecks of the model. Therefore, we utilized the TIDE toolkit for error classification and contribution analysis. TIDE categorizes detection errors into six types (classification error, localization error, mixed error, duplicate detection, false positive, and false negative) and provides a more granular analysis of error sources, as shown in Fig. 9. This section focuses on a quantitative analysis related to accuracy, with experiments conducted using ResNet-50 as the backbone, and the errors of various models were computed against the VisDrone2019 test set GT (Ground Truth).

The experimental results, as shown in Table 3, provide a detailed analysis of the six common error types in object detection tasks. The proposed HCTD-R50 demonstrates excellent performance in classification error, with a classification error of 13.91, outperforming the best one-stage detector, ATSS (14.4), and the two-stage detector, Faster-RCNN (14.43). It also significantly outperforms similar detectors such as DINO-DETR (15.89) and RT-DETR (21.37), indicating stronger performance in object category discrimination and more efficient feature extraction. In terms of localization error, HCTD (3.91) is slightly lower than Faster-RCNN (3.96), achieving the smallest error. Compared to the best one-stage detector, YOLOv10-X (4.20), and the best-performing

Table 2
Performance comparison with other feature pyramid networks.

Method	Backbone	Params(M)	AP ₅₀ (%)	AP ₇₅ (%)	AP _S (%)	AP _M (%)	AP _L (%)
RT-DETR (Zhao et al., 2024)	R-18	20.1	36.5	20.9	11.6	31.0	31.3
FPN (Lin et al., 2017)	R-18	20.7	35.4	18.8	10.8	30.4	31.1
PAFPN (Liu et al., 2018)	R-18	23.3	36.1	19.7	9.9	31.0	31.7
BiFPN (Tan et al., 2020)	R-18	21.9	36.9	20.9	11.5	31.7	34.2
SlimNeck (Li et al., 2024a)	R-18	21.1	36.4	21.0	11.9	32.0	31.4
AFPN (Yang et al., 2023)	R-18	19.7	37.1	21.0	11.7	31.8	32.8
CFPT (Du et al., 2024)	R-18	22.0	37.3	22.1	12.2	33.1	35.4
RCFPN (Zong et al., 2021)	R-18	24.4	36.8	21.5	11.1	31.4	33.6
SSFPN (Hong et al., 2021)	R-18	25.3	37.2	21.7	12.1	32.7	35.2
AugFPN (Guo et al., 2020)	R-18	21.5	36.5	18.5	10.4	29.9	33.0
GC2FPN	R-18	21.6	37.4	22.2	12.3	32.9	35.7
RT-DETR (Zhao et al., 2024)	R-50	42.3	38.5	22.4	12.7	32.5	43.4
FPN (Lin et al., 2017)	R-50	37.7	38.1	21.8	10.9	31.8	42.7
PAFPN (Liu et al., 2018)	R-50	40.3	38.4	22.1	11.1	32.1	43.6
BiFPN (Tan et al., 2020)	R-50	37.4	39.0	22.4	12.2	33.3	44.5
SlimNeck (Li et al., 2024a)	R-50	38.1	38.6	22.0	12.8	34.2	43.9
AFPN (Yang et al., 2023)	R-50	45.3	39.2	22.2	13.0	33.9	44.4
CFPT (Du et al., 2024)	R-50	38.9	40.7	23.0	13.6	35.9	45.2
RCFPN (Zong et al., 2021)	R-50	37.6	39.1	22.3	11.9	32.7	44.9
SSFPN (Hong et al., 2021)	R-50	42.4	39.9	23.1	13.3	35.4	45.3
AugFPN (Guo et al., 2020)	R-50	39.5	38.7	22.5	12.4	32.6	44.7
GC2FPN	R-50	43.9	40.9	23.2	13.8	35.6	45.8

Table 3
Error distribution of different models.

Method	Cls(%)	Loc(%)	Cls+Loc(%)	Bkgd(%)	Duplicate(%)	Missed(%)
YOLOv5-L	17.83	5.91	0.66	2.81	0.21	15.14
YOLOv8-L	15.98	4.36	0.55	2.55	0.10	14.41
YOLOv10-X	16.49	4.20	0.60	2.44	0.19	14.79
ATSS	14.40	4.41	0.58	2.75	0.08	12.80
RetinaNet	14.55	5.29	0.44	1.65	0.14	14.42
TOOD	15.49	4.58	0.52	2.18	0.07	13.81
Faster-RCNN	14.43	3.96	0.55	2.03	0.03	19.52
DINO-DETR	15.89	4.53	0.75	2.77	0.17	13.54
RT-DETR	21.37	4.18	0.55	2.37	0.11	12.62
HCTD	13.91	3.91	0.41	1.91	0.09	12.09

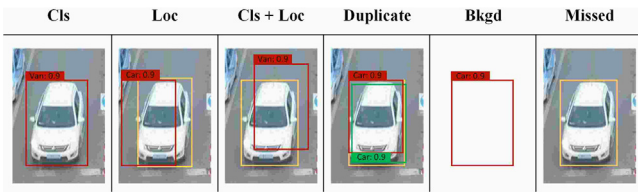


Fig. 9. Six types of errors, with red boxes representing false positive detections, yellow boxes representing ground truth, and green boxes representing true positive detections. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

DETR detector, RT-DETR (4.18), HCTD reduces the error by 6.9% and 6.4%, respectively. HCTD also performs well in regression tasks. Regarding mixed classification and localization errors, most models show errors within 1%, while HCTD achieves a low error of 0.41%, comparable to existing state-of-the-art models. The error for duplicate detection is minimal, with HCTD at only 0.09%. For background false positive errors, HCTD achieves an error of 1.91%, second only to RetinaNet (1.65%), outperforming other models, which indicates strong robustness to background interference, though further optimization is still possible. In terms of missed detection, HCTD performs the best, significantly outperforming Faster-RCNN (19.52%) and YOLOv5-L (15.14%), showing higher recall rates for objects and more stable performance in UAV aerial image detection tasks. Overall, HCTD demonstrates significant advantages in classification accuracy, localization accuracy, and object recall rate, with stable performance in both regression and classification tasks. Through TIDE's granular analysis, we have identified HCTD's strengths and potential optimization directions, providing

insights for further improving the model's performance. As shown in Fig. 10, we present a visual comparison of the absolute and relative contributions of the six error types.

4.6. Ablation study

To evaluate the contributions of each proposed component, we conducted a series of ablation experiments by incrementally integrating the FFM, CASFI, GC2FPN, and SFE into the baseline model (RT-DETR with ResNet-50 backbone, without FPN). The results are summarized in Table 4. Introducing FFM alone leads to a noticeable improvement in performance, with AP₅₀, AP₇₅, and AP_S increasing by 4.42%, 3.12%, and 3.93%, respectively, demonstrating the module's effectiveness in suppressing redundant features and enhancing semantic representation. However, when CASFI is introduced independently (row 2), a slight drop in performance is observed compared to the baseline (AP₅₀: 37.8% vs. 38.5%). This decrease can be attributed to the replacement of the global self-attention mechanism with a more localized convolutional additive attention. Unlike traditional Transformers, CASFI models token interactions via depthwise separable convolution and spatial/channel attention, which, while efficient, may sacrifice global dependency modeling. Nevertheless, the FPS increases from 98 to 112, highlighting CASFI's advantage in computational efficiency. When combining FFM and GC2FPN (row 3), the model achieves substantial improvements across all metrics (AP₅₀: 42.1%, AP_S: 13.9%), confirming the effectiveness of multi-scale global context fusion in enhancing small object detection. Adding CASFI alongside FFM and GC2FPN (row 4) slightly improves performance compared to row 3 and results in a good balance between accuracy and speed (FPS = 90). This suggests that CASFI complements other modules when used in combination.

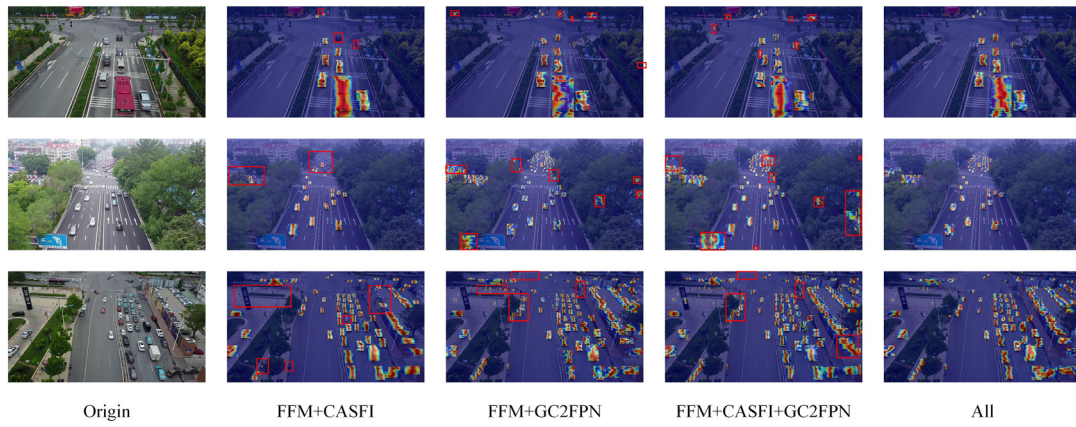


Fig. 10. Visualization of ablation experiments. Areas with significant differences are marked with red boxes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4

Ablation study of the proposed methods.

FFM	CASFI	GC2FPN	SFE	AP ₅₀ (%)	AP ₇₅ (%)	AP _S (%)	Params(M)	FPS _{bs=1}
Baseline				38.5	22.4	12.7	42.3	98
✓	×	×	×	40.2	23.1	13.2	42.9	94
×	✓	×	×	37.8	22.0	12.1	42.1	112
✓	×	✓	×	42.1	23.4	13.9	44.0	89
✓	✓	✓	×	41.6	23.3	13.8	43.7	90
✓	✓	✓	✓	42.7	23.9	14.2	44.4	85

Finally, incorporating the full set of modules including SFE (row 5) yields the best performance (AP₅₀: 42.7%, AP_S: 14.2%) but at the cost of a modest drop in speed (FPS = 85). While SFE only slightly improves accuracy, it enhances architectural scalability and can be beneficial in more complex multi-task scenarios. Overall, the ablation results demonstrate that each module contributes to performance in a complementary manner, with FFM and GC2FPN providing the most significant accuracy gains, and CASFI offering computational benefits through efficient token interaction modeling. To more clearly demonstrate the effects of the improved modules, Fig. 10 provides heatmaps generated by the prediction heads of the ablation experiment models, selected from three scenarios with varying crowding levels.

5. Discussion

Our method focuses on addressing the challenge of multi-scale object detection in UAV aerial images. Although the aforementioned analysis highlights the advantages of the proposed approach, there remain critical limitations that need to be addressed in future work. By leveraging high-performing feature selection modules and feature pyramid structures, our method processes input images to retain key information of various objects as much as possible, thereby mitigating the difficulties of multi-scale detection in UAV aerial images. However, HCTD still struggles in scenarios involving extremely small objects and densely overlapping objects, where it fails to accurately classify tiny objects and count objects in overlapping regions. While we adopted a single-stage detection paradigm to build our model, future work could draw inspiration from the global-to-local processing of two-stage detection frameworks to enhance the feature encoder and produce higher-quality feature maps for subsequent processing. Furthermore, it is essential to avoid overly complex network structures that may lead to excessive computational costs, ensuring the model can be efficiently deployed on resource-constrained UAV platforms.

6. Conclusion

This paper introduces an end-to-end hybrid detector, HCTD, based on CNN and Transformer, specifically designed for object detection in

UAV aerial imagery. HCTD effectively refines the features extracted by the backbone network, mitigating the impact of redundant image noise on model performance. It is particularly well-suited for object detection tasks in aerial images, where key information is inherently scarce. Specifically, HCTD consists of a carefully designed Feature Filtering Module (FFM), a Global Semantic Feature Flow Pyramid Network (GC2FPN), and a Shift Feature Enhancement module (SFE). The FFM and GC2FPN work collaboratively to fully integrate multi-level feature maps, endowing the model with robust global modeling capabilities, which are crucial for detecting small objects. Finally, SFE reconstructs the fused multi-level features, enhancing the non-linear interactions between features while reducing the computational overhead associated with conventional linear layers or 1×1 convolutions combined with activation functions. Extensive experiments on the VisDrone2019 aerial dataset demonstrate that HCTD outperforms other state-of-the-art detectors while maintaining high real-time detection performance.

CRedit authorship contribution statement

Hongcheng Xue: Writing – original draft, Software, Methodology, Investigation. **Zhan Tang:** Validation, Formal analysis. **Yuantian Xia:** Writing – review & editing, Validation, Supervision. **Longhe Wang:** Writing – review & editing, Visualization. **Lin Li:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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